Participant Profile Form

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organisation (SMEs, NGOs, patient organisations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centered on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	Universidad de Valladolid
Short name / acronym used in projects	UVA
Participant Identification Code (PIC)	999862809
Website(s)	www.ioba.es
Country	Spain
Organisation type	X Academic ☐ Non-academic ☐ Hospital / clinic ☐ SME ☐ Large company ☐ Research infrastructure ☐ NGO / foundation ☐ Public body ☐ Other
Main address	Paseo de Belén 17, 47011, Valladolid, España
Department(s), institute(s), research group(s) or unit(s) that would be involved	Retina Group, IOBA, University of Valladolid
Additional institutional resources relevant to collaboration	Facilities and equipment's of the IOBA-UVa (https://www.ioba.es/investigacion/laboratorios-y-unidades-de-investigacion-clinica/) *and* centralized services of UVa (https://aplicaciones.educa.jcyl.es/infrared/welcome.action).

Main contact person for this profile	Ivan Fernández Bueno, Associate Professor, Ophthalmology, UVa
Main contact email	Ivan.fernandez.bueno@uva.es
Other contact person(s), if relevant	Yolanda Álvarez, yolanda.alvarez.teruel@uva.es (Project Manager, Unidad de Excelencia IOBA)

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)

☐ Ocular surface	X Retina
X Inherited retinal dystrophies (IRDs)	☐ Age-related ocular disease
☐ Neuro-vision / integrated vision / neuro-ophthalmology	☐ Optometry / visual behaviour / rehabilitation
X Advanced therapies / regenerative medicine	X Gene therapy / cell therapy / tissue engineering
X Inflammation / biomarkers / neuroprotection	☐ Drug delivery / formulations / nanomedicine
☐ Drug discovery / screening / repurposing	☐ Ocular vascular disease
☐ Infection / trauma-related ocular disease	☐ Ocular pain / quality of life / patient-reported outcomes
☐ Health technology assessment	X AI-assisted diagnosis / software / digital tools
☐ Clinical trials / translational validation	X Other (please specify): Biocompatibility of medical devices

Free Keywords for internal searchable database (please provide up to 10)	iPSC; retinal organoids; iRPE; animal models; preclinical models; neurodegeneration; biocompatibility; medical devices; neuroretina culture
Geographies of interest for exchanges (optional)	Germany; Ireland; New Zealand; Sweden

Overview of your research / innovation focus

Our research and innovation focus centres on advancing the understanding and treatment of inherited retinal dystrophies (IRDs) through a comprehensive preclinical approach. By investigating loss-of-function mechanisms associated with specific genetic mutations, we aim to clarify their phenotypic variability and underlying pathophysiology, acknowledging the complexity whereby different mutations may lead to similar clinical presentations and, conversely, the same mutation may produce divergent outcomes.

This work leverages patient-derived induced pluripotent stem cells (iPSCs), retinal organoids and gene-editing technologies such as CRISPR-Cas9 to correct mutations, model disease and dissect molecular mechanisms. Current efforts include the in vitro and in vivo characterisation of IRDs linked to the c.1354dupT mutation in the PROM1 gene, using iRPE models and knock-in mouse systems, with a strong emphasis on rare diseases and translational applicability.

In parallel, we explore neuroprotective strategies aimed at preserving retinal structure and function, including repurposing existing drugs to mitigate neurodegeneration, while also contributing to the preclinical biocompatibility assessment of ophthalmic medical devices under ISO 10993 standards. This combined approach supports the safe and effective translation of novel therapies and devices into clinical practice through the evaluation of cytotoxicity, irritation, sensitisation and toxicity.

Complementing these experimental and translational activities, the team currently co-leads the national multi-centre project StargSpain (Precision Medicine and Advanced Diagnosis in Stargardt Disease in Spain), focused on the phenotypic and genotypic characterisation of patients with ABCA4-associated retinopathies using whole genome sequencing and multimodal retinal imaging. The project also incorporates Artificial Intelligence and Deep Learning approaches to develop advanced diagnostic and predictive tools based on OCT and fundus autofluorescence, strengthening the integration of clinical datasets with our preclinical research lines..

Why your team could be relevant to a translational eye-health and vision consortium

Our team brings distinctive added value to a consortium focused on visual health and translational ophthalmology through an integrated, end-to-end preclinical capability that bridges fundamental research and clinical application. We contribute deep expertise in modelling inherited retinal dystrophies using patient-derived iPSC systems and advanced gene-editing tools such as CRISPR-Cas9, enabling precise investigation of disease mechanisms and validation of therapeutic strategies in a human-relevant context. Our ongoing work on PROM1-associated retinal degeneration, combining in vitro retinal organoids and iRPE models with in vivo knock-in mouse systems, positions us to generate robust, translatable data for rare and genetically complex conditions.

In parallel, our research on neuroprotection provides a complementary therapeutic dimension, focused on preserving retinal structure and function across degenerative diseases, including through drug-repurposing approaches that can accelerate clinical translation. This dual focus on disease modelling and therapeutic intervention strengthens the consortium's capacity to move from mechanistic insight to actionable treatment strategies.

Additionally, our dedicated platform for preclinical biocompatibility assessment of ophthalmic drugs and medical devices, aligned with ISO 10993 standards, offers critical infrastructure for safety evaluation. This enables early identification of cytotoxicity, irritation, sensitisation and systemic toxicity risks, facilitating regulatory readiness and de-risking innovation pipelines.

Our team also contributes expertise in multimodal retinal imaging, genotype–phenotype correlation studies, and the application of Artificial Intelligence and Deep Learning tools for advanced diagnosis and prediction in inherited retinal diseases. This clinical and computational dimension reinforces our ability to integrate robust patient-oriented datasets with preclinical research, supporting precision medicine strategies and clinically meaningful translational outcomes. Altogether, our team enhances the consortium's ability to deliver scientifically rigorous, clinically relevant and translationally impactful outcomes.

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute

- **Advanced therapy development pipeline** spanning neuroprotection and regenerative medicine approaches for retinal diseases, with a strong orientation toward clinically actionable outcomes, supported by extensive experience in preclinical development and validation, as well as clinical translation through investigator-initiated trials in cell-based therapies for retinal disorders, and gene therapy strategies leveraging CRISPR-Cas technologies.
- **Organotypic neural retina culture platform** enabling ex vivo evaluation of neuroprotective compounds, providing a physiologically relevant intermediate between cell culture and in vivo models.
- **Custom preclinical model generation** (both in vitro and in vivo) tailored to specific retinal pathologies, supporting efficacy testing and mechanism-of-action studies for emerging therapies.
- **Expertise in iPSC technology and retinal organoids**, allowing the study of human disease biology in complex, patient-specific systems and facilitating therapeutic screening.
- **Multimodal retinal imaging and genotype–phenotype correlation approaches,** integrating whole genome sequencing (WGS), OCT, fundus autofluorescence and Artificial Intelligence/Deep Learning methodologies for advanced diagnosis, prediction and precision medicine strategies in inherited retinal diseases.
- **Integrated innovation ecosystem**, including leadership in competitive research projects, active collaboration with industry partners, and co-invention of patents that drive technology transfer and real-world application.
- **Translational acceleration capacity**, combining experimental validation with strategic alignment toward regulatory and clinical development pathways.
- **Specialised platform for ophthalmic device safety assessment**, supporting early-stage validation and risk mitigation in parallel with therapeutic development.
- **Training and knowledge transfer hub**, with the ability to host and train visiting researchers in advanced retinal models and techniques, as well as to send team members abroad to acquire and implement cutting-edge methodologies.
- **Collaborative flexibility**, adapting experimental systems and expertise to consortium needs, from mechanistic studies to preclinical proof-of-concept and technology validation.

Infrastructure, equipment, facilities or datasets relevant to consortium work

Facilities and equipment's of the IOBA-UVa (*https://www.ioba.es/investigacion/laboratorios-y-unidades-de-investigacion-clinica/*):

- *Cell Culture Lab*: equipped with laminar flow hoods, incubators, ovens, water baths, stereo, light and epifluorescence microscopes, electroporator, cell counters, transfectors, centrifuges, liquid nitrogen tanks, -80°C and -20°C freezers, among others.
- *Ocular Pathology Lab*: equipped with automated tissue processors, paraffin embedding unit, microtomes, cryostat, routine staining batteries, and epifluorescence microscope.
- *Molecular Biology Lab*: equipped with PCR and RT-PCR equipment, laminar flow hood, centrifuges, GenePix, microplate reader, gel electrophoresis system, and sonicator, anmong others.
- *Shared Labs*: equipped with autoclaves, laminar flow hoods, ovens, precision balances, pipettes, ice machines, sterilization equipment, centrifuges, and pHmeters, among others.
- *Preclinical and translational platforms:* established infrastructure for ex vivo organotypic neural retina culture, in vitro retinal disease models, and in vivo animal models for efficacy, safety and neuroprotection studies.
- *Clinical and multimodal imaging datasets:* access to phenotypic, genotypic and multimodal retinal imaging datasets associated with inherited retinal diseases, including OCT and fundus autofluorescence data integrated with whole genome sequencing and Artificial Intelligence/Deep Learning analytical approaches.

Through the centralized services of Uva, access is available to:

- *Instrumental Techniques Laboratory (LTI-INFRARED):* for confocal microscopy and flow cytometry studies (LTI-INFRARED Platform, https://aplicaciones.educa.jcyl.es/infrared/welcome.action).
- *Animal Research and Welfare Service (SIBA):* providing specialised facilities for housing, breeding, monitoring and experimental procedures involving animal models.

Training capacity and knowledge-transfer potential

Inbound staff in our organisation would benefit from a highly interdisciplinary training environment, with a primary focus on scientific, technical, clinical, computational and translational skills related to retinal diseases and visual health research. They would gain hands-on experience in advanced retinal disease modelling, including iPSC-derived systems, retinal organoids, iRPE models, and the development and application of tailored in vitro and in vivo preclinical models.

From a technical perspective, visitors would be trained in cutting-edge methodologies such as gene editing, including CRISPR-Cas-based approaches, as well as experimental design for the evaluation of novel therapeutic strategies. A key added value is the opportunity to test and validate new compounds—either developed within their home institutions or generated through collaborative projects—using our established preclinical platforms, particularly our ex vivo organotypic neural retina culture model, in which we have extensive expertise and a strong track record of robust and reproducible neuroprotection studies. Training would also include preparation, maintenance and application of these experimental systems, together with downstream cellular, molecular and functional analyses.

In addition, inbound researchers could gain exposure to multimodal retinal imaging analyses, genotype–phenotype correlation studies, and the integration of Artificial Intelligence and Deep Learning tools applied to inherited retinal diseases, including OCT and fundus autofluorescence-based approaches linked to precision medicine strategies.

On the translational side, staff would gain insight into how preclinical and clinical-oriented findings are structured and validated to support downstream application, including alignment with clinically meaningful endpoints and regulatory considerations. Overall, the training environment provides a comprehensive framework for advancing innovative therapeutic and diagnostic approaches in retinal diseases from early-stage research towards translational impact.

What your team would hope to gain from participating

Our team aims to gain access to complementary expertise, advanced technologies and innovative translational approaches that can strengthen and expand our current research lines in retinal diseases and visual health. In particular, we are interested in establishing collaborations with groups working on novel neuroprotective compounds, advanced gene and cell therapy strategies, multimodal imaging, Artificial Intelligence applications, and emerging analytical platforms that could enhance the depth and translational relevance of our experimental findings.

A central objective is to build long-term international scientific collaborations that extend beyond short-term exchanges and evolve into sustainable partnerships supported by joint projects, coordinated funding applications, scientific communications and high-impact publications. We are especially interested in integrating complementary datasets, technologies and methodological approaches that improve the robustness, reproducibility and clinical extrapolation of preclinical and patient-oriented results.

We strongly support the mobility and international training of PhD students, who would benefit from exposure to new research environments, experimental methodologies, institutional cultures and multidisciplinary working dynamics. These experiences are expected to significantly enrich their scientific and professional development, fostering adaptability, independence, international networking and broader perspectives on translational research.

At the same time, senior researchers and technical staff participating in exchanges would have the opportunity to acquire and implement novel methodologies, analytical tools and experimental strategies that could be strategically incorporated into our laboratories and research workflows. This includes exposure to infrastructures and technologies not currently available in our institution, as well as the possibility of refining key translational and methodological aspects to ensure more reliable and clinically meaningful interpretation of results.

Overall, participation in the consortium is expected to promote a dynamic and bidirectional transfer of knowledge, strengthening scientific excellence, innovation capacity and long-term collaborative impact across all participating groups.

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
Iván Fernández Bueno	Associate Professor and Principal Investigator	IOBA	Researcher	Experienced	ivan.fernandez.bueno@uva.es
Rosa Coco Martín	Full Professor and Principal Investigator	IOBA	Researcher	Experienced	rosamcoco@uva.es
Yenisey García Ferrer	PhD candidate	IOBA	Researcher	Early-Stage	yenisey.garcia22@uva.es
Yolanda Álvarez Teruel	Project Manager	IOBA	Managerial	Experienced	Yolanda.alvarez.teruel@uva.es
TBD	2 x Predoctoral Students	IOBA	Researcher	Early-Stage	

Main disciplines represented in your team	Cell Biology, Molecular Biology, Genetics and Genomics, Experimental and Translational Ophthalmology, Retinal Disease Modelling, Regenerative Medicine, Neuroprotection, Gene Therapy, Preclinical and Bioengineering Models, Artificial Intelligence and Deep Learning applied to retinal imaging, Precision Medicine, and Multimodal Retinal Imaging Analysis.
Experience in EU / international / multi-partner collaborative projects	Participation as an early-stage researcher in a secondment within the *Drug Discovery and Development of Novel Eye Therapeutics (3D-NET)* project (FP7-PEOPLE-2013-IAPP), where I also designed, conducted, and led experimental work supervising seconded staff at my home institution (UVA/IOBA).
Experience with clinical, translational or patient-facing research (if relevant)	☐ Yes X No ☐ Not applicable | **Short note:** *Translational/preclinical retinal research; no direct patient-facing clinical activity indicated in this profile.*

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements

1. *Puertas-Neyra K, Coco-Martin RM, Hernandez-Rodriguez LA, Gobelli D, Garcia-Ferrer Y, Palma-Vecino R, Tellería JJ, Simarro M, de la Fuente MA, Fernandez-Bueno I.* ***Clinical exome analysis and targeted gene repair of the c.1354dupT variant in iPSC lines from patients with PROM1-related retinopathies exhibiting diverse phenotypes****. Stem Cell Research & Therapy 2024;15:192. DOI: 10.1186/s13287-024-03804-2*
2. *Usategui-Martín R, Puertas-Neyra K, Galindo-Cabello N, Hernández-Rodríguez LA, González-Pérez F, Rodríguez-Cabello JC, González-Sarmiento R, Pastor JC, Fernandez-Bueno I*.* ***Retinal Neuroprotective Effect of Mesenchymal Stem Cells Secretome Through Modulation of Oxidative Stress, Autophagy, and Programmed Cell Death****. Investigative Ophthalmology & Visual Sciences 2022;63:27. DOI: 10.1167/iovs.63.4.27*
3. *Usategui-Martín R, Puertas-Neyra K, García-Gutierrez MT, Fuentes M, Pastor JC, Fernandez-Bueno I*.* ***Human mesenchymal stem cell secretome exhibits a neuroprotective effect over in vitro retinal photoreceptor degeneration****. Molecular Therapy - Methods & Clinical Development 2020;17:1155-1166. DOI: 10.1016/j.omtm.2020.05.003*
4. *Srivastava GK, Alonso-Alonso ML, Fernandez-Bueno I, Garcia-Gutierrez MT, Rull F, Medina J, Coco RM, Pastor JC.* ***Comparison between direct contact and extract exposure methods for PFO cytotoxicity evaluation****. Scientific Reports 2018;8:1425. DOI: 10.1038/s41598-018-19428-5*
5. *Fernandez-Bueno I, Fernandez-Sanchez L, Gayoso MJ, Garcia-Gutierrez MT, Pastor JC, Cuenca N.* ***Time course modifications in organotypic culture of human neuroretina****. Experimental Eye Research 2012; 104:26-38. DOI: 10.1016/j.exer.2012.08.012*

Up to 5 relevant previous projects, grants, networks or collaborations

Design and characterization of an in vitro model of Retinitis Pigmentosa due to a mutarion on the PROM1 gene by using genetic edition of induced Pluripotent Stem Cells (RP-PROM1)
Diseño y caracterización de un modelo in vitro de Retinitis Pigmentosa por mutación en el gen PROM1 mediante edición genética de Células Madre Pluripotenciales inducidas (RP-PROM1)

- *Proyectos de I+D+i – Generación de Conocimiento, Convocatoria 2020, Ministerio de Ciencia e Innovación.*
- *Referencia: PID2020-118860RB-I00*
- *Instituto Universitario de Oftalmobiología Aplicada (IOBA), Universidad de Valladolid.*
- *Coco Martín RM (Investigador Principal 1), Fernández Bueno I (Investigador Principal 2), Srivastava GK (Equipo de Trabajo), Tellería Orriols JJ (Equipo de Trabajo), García Gutiérrez MT (Equipo de Trabajo).*
- *Importe: 105.335€*
- *4 años, 01/09/2021-31/08/2025.*

Design, Development and Validation of a Highly Purified Perfluorooctane for Use in Ophthalmic Surgery: JUSTPERFLUORO
Diseño, desarrollo y validación de un perfluroooctano altamente purificado para uso en cirugía oftalmológica: JUSTPERFLUORO

- *Proyectos de colaboración público-privada 2021, Ministerio de Ciencia e Innovación.*
- *Referencia: CPP2021-008695*
- *AJL Opththalmic S.A.; VISION I MAS D S.L.; y Universidad de Valladolid (Instituto de Oftalmobiología Aplicada -IOBA- y Grupo de Tecnologías a Presión -PressTech-)*
- *AJL Opththalmic S.A.: Larra E (Coordinadora), Barrio P (Investigador), Cobo MT (Investigador), Uriarte N (Investigador), Martínez A (Investigador), Gondra Vázquez A (Investigador), Peña Aldrey CA (Investigador); VISION I MAS D S.L.: García Gutiérrez MT (Investigador); Universidad de Valladolid: Fernández Bueno I (Investigador Principal), García Serna J (Investigador), Cantero Sposetti DA (Investigador) y Pastor Jimeno JC (Investigador).*
- *Importe: 925.208€ (AJL Opththalmic S.A.: 194.974,11€; VISION I MAS D S.L.: 147.978,60€; Universidad de Valladolid: 110.784,00€)*
- *3 años, 01/03/2022-31/12/2025.*

In Vitro and In vivo Modeling and characterization of Inherited Retinal Diseases due to the c.1354dupT mutation in PROM1, using iRPE and murine knock-in models (IVIM-PROM1).
Modelado y caracterización in vitro y in vivo de las Distrofias Hereditarias de la Retina originadas por la mutación c.1354dupT en PROM1, mediante modelos iRPE y ratones knock-in (IVIM-PROM1)

- *Proyectos de Generación de Conocimiento, Convocatoria 2024, Ministerio de Ciencia, Innovación y Universidades*

- *Referencia: PID2024-160341OB-I00*
- *Instituto Universitario de Oftalmobiología Aplicada (IOBA), Universidad de Valladolid.*
- *Fernández Bueno I (Investigador Principal 1), Coco Martín RM (Investigador Principal 2), Tellería Orriols JJ (Equipo de Investigación) , Cuadrado Asensio R (Equipo de Investigación), Puertas Neyra KL (Equipo de Trabajo), Hernández Rodríguez LA (Equipo de Trabajo), Andrés Iglesias C (Equipo de Trabajo), Bravo Nuevo A (Equipo de Trabajo), García Vázquez C (Equipo de Trabajo)*
- *Importe: 141.250€*
- *4 años, 01/09/2025-31/08/2029.*

Ocular Topical administration of Sorbitan ester Nanoparticles for Intraocular Neuroprotection (RETINARTEX)
Administración tópica ocular de nanopartículas de ésteres de sorbitán para neuroprotección intraocular (RETINARTEX)

- *Ayudas a Proyectos de "Prueba de Concepto" 2022, Ministerio de Ciencia e Innovación*
- *Referencia: PDC2022-133239-C21*
- *Instituto Universitario de Oftalmobiología Aplicada (IOBA), Universidad de Valladolid*
- *Diebold Luque Y (Investigador Principal 1), García Posadas L (IP 2), Fernández Bueno I (Investigador Colaborador), Puertas Neyra KL (IC), Romero Castillo I (IC), Ossa Calderón C (IC), Andrés Iglesias C (IC), Sánchez Barreiro A (IC).*
- *Importe: 115.000€*
- *2 años, 01/12/2022-30/11/2024*

Precision Medicine and Advanced Diagnosis in Stargardt Disease in Spain: Diagnostic Tools through the Use of Genetic Biomarkers, Imaging, and Artificial Intelligence Analysis. Acronym: StarSpain
Medicina de precisión y diagnóstico avanzado en la Enfermedad de Stargardt en España: Herramientas Diagnósticas mediante el Uso de Biomarcadores Genéticos, de Imagen y Análisis con Inteligencia Artificial (StargSpain)

- *Proyectos de Investigación en Enfermedades Raras 2024, Misiones Conjuntas del Ministerio de Sanidad y del Ministerio de Ciencia, Innovación y Universidades, Instituto de Salud Carlos III*
- *Referencia: PMPER24/00022*
- *Fundación Instituto de Investigación Sanitaria de Aragón, Instituto de Investigación Sanitaria de Aragón, Universidad de Zaragoza*
- *Grupos Participantes: Instituto de Investigación Sanitaria de Aragón, Universidad de Valladolid, Hospital de la Paz, Biodonostia, Instituto oftalmológico Fernández-Vega, Hospital Virgen Macarena, Universidad de Barcelona, Hospital La Fé, Universidad de Navarra*
- *Pinilla Lozano I (IP), Coco Martín RM (IP Corresponsable), Fernández Bueno I (Investigador Colaborador UVa), Sánchez Tocino H (IC UVa), Cuadrado Asensio R (IC UVa), Andrés Iglesias C (IC UVa), Blázquez Araúzo F (IC UVa)*
- *Importe: 1.167.650€ (UVa: 162.341,69€)*
- *2 años, 01/01/2025-31/12/2026*

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)

Pharmaceutical Composition for the Treatment and/or Prevention of Degenerative Retinal Diseases
Composición farmacéutica para el tratamiento y/o prevención de enfermedades retinianas degenerativas

- *Inventores: Srivastava GK, Singh AK, Fernández Bueno I, Gayoso MJ, Pastor Jimeno JC*
- *Número de patente o CCP: ES 2539329 A1*
- *Fecha de concesión: 29/06/2015*
- *Número de la solicitud: P201331743*
- *Fecha de solicitud: 29/11/2013*
- *Número de patente internacional: WO/2015/079093*
- *Fecha de concesión: 04/06/2015*
- *Número de la solicitud internacional: PCT/ES2014/070879*
- *Clasificación Internacional de Patentes: C12N 5/0775 (2010.01)*
- *País de prioridad: España*
- *Entidad titular: Universidad de Valladolid*

Method for Evaluating the Cytotoxicity of Chemical Substances
Método para evaluar la citotoxicidad de sustancias químicas

- *Inventores: Srivastava GK, Alonso Alonso ML, Fernández Bueno I, García Gutiérrez MT, Coco Martín RM, Pastor Jimeno, JC.*
- *Número de patente o CCP: ES2644987*
- *Número de patente europea: EP2080.12-PACE*
- *Fecha de concesión: 05/10/2018*
- *Número de la solicitud: P201630708*
- *Fecha de solicitud: 30/05/2016*
- *Número de la solicitud internacional: PCT/ES2017/070365*
- *Fecha de solicitud: 29/05/2017*
- *Número de publicación: EP3467118*
- *Fecha de publicación: 10/04/2019*
- *País de prioridad: España*
- *Entidad titular: Universidad de Valladolid*

- *Validación externa: "External validation of a medical devices cytotoxicity testing method: IOBA-CYTOTEST". Final Report Study Code: 4465 14 november 2019*
- *Empresa/s que la están explotando: Vision I+D S.L., Valladolid, España*

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	X Yes ☐ Maybe ☐ No capacity to send staff on secondment
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	X Only Researchers ☐ Technical Staff/Innovation X Managerial / project support ☐ Admin Staff ☐ Other profiles
Could your organisation host incoming staff from academic and/or non-academic sectors?	☐ Academic only ☐ Non-academic only X Both ☐ No capacity to host
Administrative / HR / legal considerations that may affect mobility	None

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice

- **Organotypic neural retina culture platform** as a core strength for the **direct evaluation of novel molecules and products**, combining hands-on experimental workflows with robust readouts through cellular and molecular biology techniques, as well as functional assays; including full training in the preparation, handling, and application of this ex vivo model.
- **Advanced human retinal modelling using iPSC technology**, including reprogramming, differentiation into retinal cell types, and generation of retinal organoids and iRPE systems for disease modelling and therapeutic testing.
- **In vivo preclinical models of retinal disease**, enabling the modelling and characterisation of pathologies and the evaluation of new therapeutic approaches in a physiologically relevant context.

Capabilities or expertise you would especially like to access through the consortium

- **Discovery and validation of neuroprotective compounds** through collaboration with complementary teams (academic and industrial), enabling the identification of novel candidates with potential relevance for retinal diseases and their evaluation across diverse preclinical platforms.
- **Advanced gene therapy approaches in vivo**, particularly partnerships that enable the implementation and optimisation of CRISPR-Cas-based strategies in animal models, with a clear focus on strengthening the pathway toward clinical translation in human retinal disorders.
- **Access to cutting-edge infrastructures and specialised technologies** not currently available in our centre, including advanced imaging, omics platforms, high-throughput screening, AI-driven analytical tools and large-scale computational approaches that could maximise the value and translational relevance of our datasets and experimental results.
- **Integration of multimodal clinical and imaging datasets**, including expertise in Artificial Intelligence, Deep Learning and genotype–phenotype correlation analyses to strengthen precision medicine approaches and predictive modelling in inherited retinal diseases.
- **Improved translational and methodological standardisation**, refining key experimental and analytical aspects that facilitate robust, reproducible and clinically meaningful extrapolation of preclinical findings.
- **Effective communication and dissemination strategies**, aiming not only to reach the scientific community through high-impact publications, but also to improve outreach to society, increasing awareness and understanding of retinal diseases and emerging therapies.
- **Long-term international collaboration frameworks**, fostering durable partnerships supported by joint research projects, coordinated funding applications, and co-authored scientific outputs.
- **Exchange of expertise and methodologies**, enabling mutual learning and integration of innovative techniques that enhance experimental capabilities across participating groups.

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	X Yes ☐ No
X We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.	

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.
None

The information provided in this form will be used only for internal consortium-building purposes in connection with the preparation of a prospective MSCA Staff Exchanges proposal (2027 call) coordinated from Unidad de Excelencia IOBA / University of Valladolid. It may be used by the coordination team to: (i) assess strategic fit and administrative readiness, (ii) prepare participant-related information required at proposal stage, and (iii) build an internal profile database to facilitate contacts among interested participants. The information will not be shared outside this pre-consortium context without prior permission from the contributing organisation / contact person, except where later required for the formal preparation of a proposal with the explicit involvement of that organisation. Submission of this form confirms that the contributor agrees to this internal use. Organisations may request correction, update or withdrawal of their information at any time by contacting the coordination team.

X Understood

Name: Ivan Fernández Bueno
Date completed: 18/5/2026

(Coordination team: ivan.fernandez.bueno@uva.es; yolanda.alvarez.teruel@uva.es)